

Yeogyeeong Song

+82-10-6338-6400 | in yeogyeeongsong | 0gonge | Blog
challengef0802@gmail.com | yg.song@lgresearch.ai

EDUCATION

Sungshin Women's University
B.S. in Artificial Intelligence

Seoul, South Korea
Feb 2021 – Feb 2026

EXPERIENCE

LG AI Research (Physical Intelligence Lab, Robotics Tech Cell)
AI Engineering Intern

Seoul, South Korea
Oct 2025 – Present

- Working on a research project on region-conditioned trajectory generation for VLA-based robotic manipulation, focusing on compositional interaction primitives, spatially grounded subtask control, and data-efficient trajectory retargeting.
- Performed real-world manipulation experiments using single-arm and dual-arm robots; built a scalable multi-modal data collection pipeline for training.

SELECTED PROJECTS

WISPY – Specialized LLM Optimization & Data Engineering
LLM & Infra Engineer

Jan 2025 – Jul 2025

- Built a 570k-record storytelling dataset via crawling and augmentation; fine-tuned Llama-2-7B using LoRA and 4-bit NF4 quantization for child-safe, personalized interactions.
- Architected a scalable MSA on AWS EKS and SageMaker using Terraform (IaC) for automated resource management, ensuring efficient model serving with CI/CD pipelines.

HEAROAD – Real-time Sign Language Recognition & Voice Visualization
AI & Backend Engineer

Mar 2024 – Dec 2024

- Developed an AI-driven accessibility service featuring real-time sign language translation and voice visualization to bridge digital gaps; awarded Gold Prize (Minister's Award, MSIT).
- Optimized inference by implementing Mixed Precision, Gradient Accumulation, and Checkpointing, reducing memory usage by 35% and enhancing speed by 30% on AWS-deployed models.

BrainWorks – MRI-based Alzheimer's Prognosis Assistant System
AI & Frontend Engineer

Mar 2024 – Dec 2024

- Developed a VGG16-based Alzheimer's diagnosis model using customized MRI slice-sampling and Grad-CAM for clinical interpretability (94.56% accuracy); awarded Bronze Prize (ACK 2024).
- Engineered a React-based clinical interface prioritizing diagnostic independence by separating physician workflows and enabling togglable AI results for unbiased decision-making.

PROJECTS

BOSINSA - Personal Color-Based Fashion Recommendation System
AI Product Engineer

Nov 2024 – Dec 2024

- Constructed a large-scale dataset of 100,000+ fashion images via crawling and optimized database performance by implementing image clustering and refined MySQL schema design for efficient retrieval.
- Architected a low-latency inference pipeline using FastAPI to bridge the AI model with a React frontend, delivering an adaptive UI that dynamically updates visual themes based on real-time personal color analysis.

iOS Development - Place Recommendation & Task Management for Procrastinators
Lead iOS Developer

Mar 2024 – Dec 2024

- Developed a video-sharing SNS utilizing MVC architecture to mitigate venue information asymmetry, recognized with a Grand Prize for technical excellence and social impact.
- Orchestrated a structural transition from MVC to MVVM architecture to enhance system scalability and decouple business logic, significantly improving code maintainability for a productivity planner.

Friendly - Generative AI-based Emotional Companion Vehicle

Mar 2024 – Sep 2024

System Integration Engineer

- Engineered an autonomous mobile platform using Raspberry Pi and a 3D-printed chassis, integrating multi-modal peripherals for holistic environment sensing.
- Developed a real-time interaction pipeline combining BERT-based sentiment analysis with the Google Assistant API to enable empathetic human-robot communication.

RPA-driven Career Guidance System with T5-based Summarization

Nov 2023

AI & RPA Developer

- Designed an autonomous workflow to aggregate and process multi-source career information, recognized with a Special Prize from the National Information Society Agency (NIA).
- Integrated a T5-based text summarization model into the RPA pipeline to synthesize unstructured qualitative reviews into structured, actionable career insights.

AI-based Crowd Dispersion & Spatial Recommendation System

Jul 2023 – Aug 2023

AI & Backend Engineer

- Implemented the FP-Growth algorithm to extract frequent mobility patterns and association rules, designing an optimized crowd dispersion strategy to mitigate spatial congestion.
- Engineered a real-time monitoring prototype utilizing computer vision to validate the system’s efficacy and situational responsiveness in high-density environments.

IoT Coaster for Drink-Spiking Prevention

Nov 2022

IoT & Server Developer

- Engineered an IoT-based real-time alert system utilizing touch sensors to prevent drink-spiking crimes, recognized with the Excellence Award for its practical social impact and technical execution.
- Resolved hardware reliability challenges by optimizing conductive interfaces for enhanced sensor sensitivity and established a robust hardware-to-mobile data pipeline for seamless notification delivery.

SKILLS

- **Programming:** Python, C++
- **Robotics & Simulation:** MuJoCo Simulator, Franka Emika, OMY, AI Worker
- **OS & Environment:** Ubuntu/Linux, Git, Docker, PyTorch
- **English:** TOEIC Speaking – Advanced Low (AL)

AWARDS

- **Excellence Award**, KB Kookmin Bank Talent Development Program (Social Contribution) *Aug 2025*
- **Gold Prize (Minister’s Award)**, ICT Probono Mentoring, Ministry of Science and ICT *Dec 2024*
- **Bronze Prize**, ACK 2024 Paper Competition, Korea Information Processing Delivery (KIPS) *Nov 2024*
- **Grand Prize**, Sopkathon, SOPT *Nov 2024*
- **Excellence Award**, BOAZ Big Data Hackathon *Nov 2024*
- **Excellence Award**, AppJam (Long-term Project), SOPT *Jul 2024*
- **Special Award**, AIRPA Hackathon, National Information Society Agency (NIA) *Nov 2023*
- **Honorable Mention**, Software Competition, Sungshin Women’s University *Aug 2023*
- **Excellence Award**, StartUp MAKER Camp, Sungshin Women’s University *Nov 2022*

EXTRACURRICULAR ACTIVITIES

- **BOAZ (Big Data Alliance Club)** *Jun 2024 – Jun 2025*
Participated in advanced AI paper seminars and achieved an Excellence Award in a data science hackathon by implementing ensemble learning and audio feature analysis.
- **SOPT (IT Venture Networking Club)** *Mar 2024 – Jan 2025*
Engaged in cross-functional team projects to build scalable iOS applications; recognized with a Grand Prize and an Excellence Award for technical innovation and product quality.
- **GDSC (Google Developer Student Clubs)** *Sep 2022 – Sep 2024*
Facilitated technical workshops and spearheaded peer-led study programs as a core member to foster a collaborative learning environment and developer mentorship.